

Week 9: Three-Phase Power

Phase and line quantities, Y and Δ loads, three-phase power, power factor correction, residential wiring, power distribution

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 8 worked examples, and 8 practice problems with full solutions.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §13.5–13.7

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

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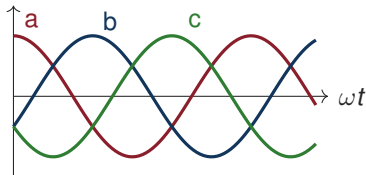
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How to use this deck

- **“Try it” slides:** the problem alone, with empty space. Do it first.
- **Next slide:** the clean solution, step by step.
- **Purple boxes:** engineering shortcuts, the fast way to see or solve things.
- **Orange boxes:** traps. This is where marks get lost on exams.
- Formulas are tagged **[on formula sheet]** or **[NOT on formula sheet]**, based on the sheet given in ELEC 275 exams.
- **End of deck:** final-check problems, then a full answer sheet.

The right side of each slide is left empty for your notes.

Why three phases?



three equal sinusoids, 120° apart

Almost all power is generated and sent as three-phase:

1. The total power delivered is **constant** (not pulsing), so motors run smoothly.
2. It needs less copper than three separate single-phase circuits for the same power.
3. Three-phase motors start and run without extra parts.

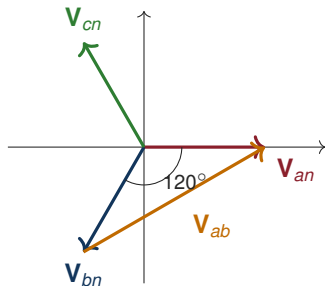
Symbols for this week

Symbol	Meaning
a, b, c	the three phases (lines). n = neutral
V_{an}	phase voltage: line a to neutral. Magnitude V_p (or V_ϕ)
V_{ab}	line voltage: line a to line b. Magnitude V_L
I_a	line current: current in wire a. Magnitude I_L
I_{ab}	phase current in a Δ load: current through the impedance between a and b
Z_Y, Z_Δ	impedance per phase of a Y load, of a Δ load
balanced	all three sources equal in size and 120° apart, all three load impedances equal

Three-phase voltages are almost always given as **RMS line voltages** (“208 V system” = $V_L = 208 \text{ V RMS}$).

Balanced source and phase sequence

[NOT on formula sheet: memorize]



Positive (abc) sequence:

$$\mathbf{V}_{an} = V_p \angle 0^\circ$$

$$\mathbf{V}_{bn} = V_p \angle -120^\circ$$

$$\mathbf{V}_{cn} = V_p \angle +120^\circ$$

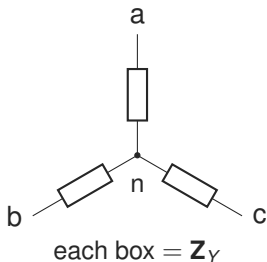
a peaks first, then b, then c.

Balanced: $\mathbf{V}_{an} + \mathbf{V}_{bn} + \mathbf{V}_{cn} = 0$.

Trap: in negative (acb) sequence, b and c swap ($\mathbf{V}_{bn} = V_p \angle +120^\circ$), which reverses a motor. Use abc unless told otherwise.

Y (wye) connection

[NOT on formula sheet: memorize]



Each impedance goes from a line to the common point n.

Line current $I_L = I_p$ (same wire)

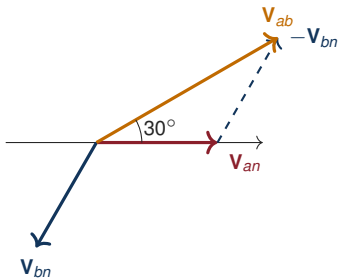
Line voltage $V_L = \sqrt{3} V_p$

Phase shift $V_{ab} = \sqrt{3} V_p \angle (\theta_a + 30^\circ)$

Engineering shortcut

Y: **voltages** get the $\sqrt{3}$ (and the 30°). Δ : **currents** get the $\sqrt{3}$. Memory aid: “Y voltage, Delta current”.

Why $V_L = \sqrt{3} V_p$ and the 30°



Line voltage = difference of two phase voltages: $V_{ab} = V_{an} - V_{bn}$.

Add V_{an} and $-V_{bn}$ tip to tail: two equal arrows 60° apart. Their sum points halfway between, at 30° , with length $2V_p \cos 30^\circ = \sqrt{3} V_p$.

Algebra check with $V_p = 1$:

$$1 - (-0.5 - j0.866) = 1.5 + j0.866 = 1.732 \angle 30^\circ \checkmark$$

The same happens for V_{bc} and V_{ca} : all three line voltages are $\sqrt{3}$ bigger and 30° ahead of their phase voltages.

Try it: Example 1, phase and line voltages

A balanced abc source has $\mathbf{V}_{an} = 120\angle 0^\circ$ V RMS.

(a) Write $\mathbf{V}_{bn}$ and $\mathbf{V}_{cn}$. (b) Find $\mathbf{V}_{ab}$ (magnitude and angle). (c) Check that the three phase voltages add to zero.

your work here ↓

Example 1: solution

(a) abc sequence: b is 120° behind a, c is 120° ahead:

$$\mathbf{V}_{bn} = 120\angle -120^\circ, \quad \mathbf{V}_{cn} = 120\angle 120^\circ$$

(b) rect $\mathbf{V}_{bn} = 120 \cos(-120^\circ) + j120 \sin(-120^\circ) = -60 - j103.9$

(b) subtract $\mathbf{V}_{ab} = \mathbf{V}_{an} - \mathbf{V}_{bn} = 120 - (-60 - j103.9) = 180 + j103.9$

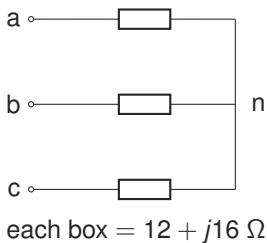
(b) polar $\sqrt{180^2 + 103.9^2} = 207.8, \tan^{-1}(103.9/180) = 30^\circ:$

$$207.8\angle 30^\circ \text{ V} = \sqrt{3} \times 120\angle (0^\circ + 30^\circ) \checkmark$$

(c) $120 + (-60 - j103.9) + (-60 + j103.9) = 0 \checkmark$ (the real parts and the imaginary parts both cancel)

This is why “120/208 V” systems exist: 120 V phase (outlets), 208 V line.

Try it: Example 2, balanced Y–Y



$V_p = 120 \, \text{V RMS (abc)}$, Y-connected load, $\mathbf{Z}_Y = 12 + j16 \, \Omega$ per phase.

Find $\mathbf{I}_a$, P , Q , $|\mathbf{S}|$ and pf.

your work here ↓

Example 2: solution (1/2)

Plan balanced Y: solve phase a alone (source V_{an} into one Z_Y), shift for b and c, multiply power by 3

$$\mathbf{Z}_Y \quad 12 + j16: \sqrt{144 + 256} = 20, \tan^{-1}(16/12) = 53.13^\circ: \\ 20\angle 53.13^\circ$$

$$\mathbf{I}_a \quad \frac{120\angle 0^\circ}{20\angle 53.13^\circ} = \boxed{6\angle -53.13^\circ \text{ A}}$$

$\mathbf{I}_b, \mathbf{I}_c$ same size, shifted -120° and $+120^\circ$: $6\angle -173.13^\circ$, $6\angle 66.87^\circ$

Example 2: solution (2/2)

P per phase $I^2 R = 36 \times 12 = 432 \text{ W}$; three phases: **1296 W**

Q per phase $I^2 X = 36 \times 16 = 576 \text{ var}$; total **1728 var**

|S|, pf $\sqrt{1296^2 + 1728^2} = \textbf{2160 VA}$, $\text{pf} = R/|Z| = 12/20 =$
0.6 lagging

Check $\sqrt{3} V_L I_L \cos \theta = \sqrt{3}(207.8)(6)(0.6) = 1296 \text{ W } \checkmark$

The per-phase method

Engineering shortcut: solve one phase, multiply by 3

In a balanced system the neutral carries **no current** ($\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 0$) and sits at 0 V, even with no neutral wire.

So: draw only phase a (source $\mathbf{V}_{an}$, one $\mathbf{Z}_Y$), solve it like a single-phase circuit, and get b and c by shifting -120° and $+120^\circ$. Total power = $3 \times$ phase power.

Line impedance goes in series with $\mathbf{Z}_Y$ in the per-phase circuit (Example 8).

Trap: the per-phase method needs a **Y** load. Convert Δ loads first: $\mathbf{Z}_Y = \mathbf{Z}_\Delta/3$.

Why the neutral carries no current

In a balanced Y load, the three currents have the same size and are 120° apart: $I\angle\alpha$, $I\angle(\alpha - 120^\circ)$, $I\angle(\alpha + 120^\circ)$.

Draw them tip to tail: three equal arrows turning 120° each time close into a triangle and end where they started, so the sum is **zero**.

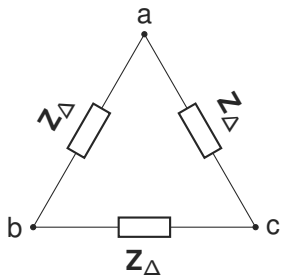
(Algebra: $1 + (-0.5 - j0.866) + (-0.5 + j0.866) = 0$.)

KCL at the load's star point: $\mathbf{I}_n = \mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 0$. With no current, the neutral wire has no drop, so the star point sits at 0 V whether or not a neutral wire is connected.

That's why each phase can be solved alone as “source $\mathbf{V}_{an}$ into $\mathbf{Z}_Y$ ”: the other two phases never affect it. Unbalanced loads break this, and then the neutral does carry current.

Δ (delta) connection

[NOT on formula sheet: memorize]



Each impedance goes line to line. No neutral.

Phase voltage $V_p = V_L$ (across a line pair)

Line current $I_L = \sqrt{3} I_p$

Phase shift $I_a = \sqrt{3} I_p \angle (\theta_{ab} - 30^\circ)$

Y- Δ conversion (balanced)

$$Z_Y = Z_\Delta / 3$$

$$Z_\Delta = 3 Z_Y$$

Why $I_L = \sqrt{3} I_p$ in Δ , and why $\mathbf{Z}_Y = \mathbf{Z}_\Delta/3$

Line current: KCL at corner a of the Δ : line current in = current out through $\mathbf{Z}_{ab}$ minus current in from $\mathbf{Z}_{ca}$: $\mathbf{I}_a = \mathbf{I}_{ab} - \mathbf{I}_{ca}$.

It's the same geometry as the Y voltages: the difference of two equal phasors 120° apart has size $\sqrt{3}$ times bigger, shifted by 30° (here lagging): $\mathbf{I}_a = \sqrt{3} I_p \angle(\theta_{ab} - 30^\circ)$.

Conversion: a Y and a Δ are equivalent if they draw the same line current from the same line voltage V_L :

$$\begin{aligned}\Delta: \quad I_L &= \sqrt{3} \frac{V_L}{|\mathbf{Z}_\Delta|} \\ Y: \quad I_L &= \frac{V_L/\sqrt{3}}{|\mathbf{Z}_Y|}\end{aligned}$$

Setting them equal: $|\mathbf{Z}_Y| = |\mathbf{Z}_\Delta|/3$, with the same angle (same power factor). So $\mathbf{Z}_Y = \mathbf{Z}_\Delta/3$.

Try it: Example 3, balanced Δ load

A Δ load with $\mathbf{Z}_{\Delta} = 30 + j40 \, \Omega$ per phase is on a 240 V RMS (line) three-phase system.

(a) Phase current (b) Line current (c) Total power (d) Check (b) with $\mathbf{Z}_Y = \mathbf{Z}_{\Delta}/3$ and the per-phase method

your work here ↓

Example 3: solution (1/2)

(a) in a Δ each impedance is connected line to line, so it sees

$$V_p = V_L = 240 \text{ V: } I_p = \frac{240}{|30 + j40|} = \frac{240}{50} = 4.8 \text{ A}$$

(b) KCL at each corner: $I_L = \sqrt{3} I_p = 1.732 \times 4.8 = 8.31 \text{ A}$

(c) power is only in the resistances: $3 \times I_p^2 R = 3 \times 4.8^2 \times 30 = 2074 \text{ W}$

Example 3: solution (2/2)

(d) convert $\mathbf{Z}_Y = \mathbf{Z}_\Delta/3 = 10 + j13.3$, $|\mathbf{Z}_Y| = 16.7 \Omega$

(d) per phase Y phase voltage $V_L/\sqrt{3} = 138.6 \text{ V}$:
 $I_L = 138.6/16.7 = 8.31 \text{ A}$ ✓ (same line current)

Three-phase power

[NOT on formula sheet: memorize]

Works for both Y and Δ (balanced):

$$P = 3V_p I_p \cos \theta = \sqrt{3} V_L I_L \cos \theta$$

$$Q = \sqrt{3} V_L I_L \sin \theta \quad |S| = \sqrt{3} V_L I_L$$

θ = angle of the **per-phase impedance** (between V_p and I_p), **not** between V_L and I_L . The total $p(t)$ is **constant**.

Trap

$\sqrt{3} V_L I_L$ uses line quantities, $3 V_p I_p$ uses phase quantities. $3 V_L I_L$ is off by $\sqrt{3}$.

Where $\sqrt{3} V_L I_L \cos \theta$ comes from

Each phase is a single-phase load, so its power is $V_p I_p \cos \theta$, and the three phases are identical: $P = 3 V_p I_p \cos \theta$.

$$\mathbf{Y}: V_p = V_L / \sqrt{3}, I_p = I_L \Rightarrow P = 3 \frac{V_L}{\sqrt{3}} I_L \cos \theta = \sqrt{3} V_L I_L \cos \theta$$

$$\Delta: V_p = V_L, I_p = I_L / \sqrt{3} \Rightarrow P = 3 V_L \frac{I_L}{\sqrt{3}} \cos \theta = \sqrt{3} V_L I_L \cos \theta$$

Same formula both ways, which is why line quantities (the only ones you can easily measure) are enough. **Why the total $p(t)$ is constant:** each phase's $p(t)$ has a $\cos(2\omega t + \dots)$ ripple, and the three ripples are evenly spaced (240° apart), so they add to zero like the currents. Only the constant P is left.

Try it: Example 4, Y load (exam style)

A balanced Y load, $\mathbf{Z}_Y = 15 + j20 \, \Omega$ per phase, is fed by a 208 V RMS (line) system.

Find I_L , the power factor, P and Q .

your work here ↓

Example 4: solution

V_p the system is quoted by line voltage; a Y phase gets
 $V_p = 208/\sqrt{3} = 120.1 \text{ V}$

I_L Y: line current = phase current
 $= V_p/|Z_Y| = 120.1/\sqrt{15^2 + 20^2} = 120.1/25 = 4.80 \text{ A}$

pf $\cos \theta = R/|Z| = 15/25 = 0.6$; inductive ($X > 0$): 0.6 lagging

P $3I^2R = 3 \times 4.80^2 \times 15 = 1038 \text{ W}$

Q $3I^2X = 3 \times 4.80^2 \times 20 = 1385 \text{ var}$

Try it: Example 5, a three-phase motor

A three-phase motor draws 20 kW at pf 0.85 lagging from a 480 V RMS (line) system.

Find the line current and the reactive power.

your work here ↓

Example 5: solution

Formula $P = \sqrt{3} V_L I_L \text{ pf}$ for the whole motor $\Rightarrow I_L = \frac{P}{\sqrt{3} V_L \text{ pf}}$

$$I_L \frac{20\,000}{\sqrt{3} \times 480 \times 0.85} = \frac{20\,000}{706.7} = \boxed{28.3 \text{ A}}$$

Q power triangle: $\theta = \cos^{-1} 0.85 = 31.8^\circ$,

$$Q = P \tan \theta = 20 \times 0.620 = \boxed{12.4 \text{ kvar}}$$

Motors are rated by their **total** three-phase power. Don't divide by 3 unless you're working per phase.

Three-phase power factor correction

[NOT on formula sheet: memorize]

Same idea as week 7: $Q_C = P(\tan \theta_1 - \tan \theta_2)$ for the whole system. Then split it over **three** capacitors.

	Voltage across each C	Each capacitor
Y-connected caps	$V_p = V_L / \sqrt{3}$	$C = (Q_C / 3) / (\omega V_p^2)$
Δ -connected caps	V_L	$C = (Q_C / 3) / (\omega V_L^2)$

Shortcut: Δ capacitors are 3 times smaller than Y ones for the same correction (they see $\sqrt{3}$ more voltage).

Try it: Example 6, correct the motor

Correct the motor of Example 5 (20 kW, pf 0.85 lagging, 480 V line, 60 Hz) to pf 0.95 lagging.

Find Q_C and the capacitance per phase for (a) Y-connected and (b) Δ -connected capacitors.

your work here ↓

Example 6: solution (1/2)

Q_C same as single-phase: $Q_C = P(\tan \theta_1 - \tan \theta_2) =$
 $20(\tan 31.8^\circ - \tan 18.2^\circ) = 20(0.620 - 0.329) =$
 5.82 kvar

Per cap three equal capacitors share it: $5820/3 = 1940 \text{ var each}$

(a) Y each Y capacitor sees the phase voltage $480/\sqrt{3} = 277.1$
V: $C = \frac{1940}{(377)(277.1^2)} = 67.0 \mu\text{F}$

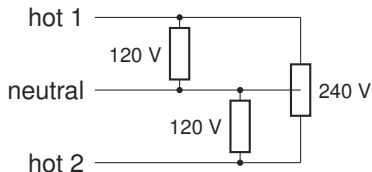
Example 6: solution (2/2)

(b) Δ each Δ capacitor sees the line voltage 480 V:

$$C = \frac{1940}{(377)(480^2)} = 22.3 \mu\text{F}$$

Why smaller Δ capacitors see $\sqrt{3}$ more voltage, and vars go as V^2 :
3 times less capacitance needed

Residential wiring (North America)



A center-tapped transformer gives two 120 V “hots” 180° apart:

hot to neutral: **120 V** (outlets, lights)

hot to hot: **240 V** (stove, dryer, AC)

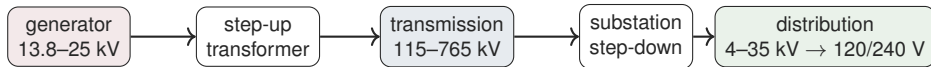
Safety

Ground wire: connects metal cases to earth, so a fault trips the breaker instead of electrifying the case.

GFCI: trips when hot and neutral currents differ by about 5 mA (current leaking through a person).

Around 10–20 mA through the body can stop you from letting go. About 100 mA across the heart can be fatal.

Power generation and distribution



Why transmit at high voltage

For the same power $P = VI$, higher V means lower I . Line loss is $I^2 R$, so 10 times the voltage means **100 times less** loss. Transformers make the voltage changes cheap, and that is why the grid is AC.

Try it: Example 7, why high voltage

A line with total resistance $5\ \Omega$ must deliver 10 MW. Treat it as single-phase for simplicity.

Find the line loss when sending at (a) 10 kV and (b) 100 kV.

your work here ↓

Example 7: solution

Idea same power P at a higher voltage needs less current ($I = P/V$), and line loss is $I^2 R$

(a) $I = 10^7/10^4 = 1000$ A, loss = $1000^2 \times 5 = 5$ MW: half the power is wasted

(b) $I = 10^7/10^5 = 100$ A, loss = $100^2 \times 5 = 50$ kW: 0.5%

Pattern 10 times the voltage, 10 times less current, $10^2 = 100$ times less loss

Try it: Example 8, line impedance

A balanced Y source, $V_p = 120$ V RMS, feeds a Y load $\mathbf{Z}_Y = 12 + j16 \Omega$ through lines with $\mathbf{Z}_{line} = 1 + j2 \Omega$ each.

Find I_L , the load phase voltage, and the power lost in the lines.

your work here ↓

Example 8: solution

Plan per-phase circuit: V_{an} drives the line impedance **in series** with Z_Y

$$\mathbf{Z} \quad (1 + j2) + (12 + j16) = 13 + j18, \\ |\mathbf{Z}| = \sqrt{169 + 324} = 22.2 \, \Omega$$

$$I_L \quad 120/22.2 = \boxed{5.40 \, \text{A}}$$

Load V_p voltage across the load impedance only:
 $5.40 \times |12 + j16| = 5.40 \times 20 = \boxed{108 \, \text{V}}$ (12 V lost on the way)

Line loss resistance of all three lines: $3 \times 5.40^2 \times 1 = \boxed{87.6 \, \text{W}}$

Traps

Check these before you box an answer

- A “480 V system” means the RMS **line** voltage.
- Y: $V_L = \sqrt{3}V_p$, $I_L = I_p$. Δ : $V_L = V_p$, $I_L = \sqrt{3}I_p$.
- θ in the pf is the per-phase impedance angle, not the line voltage-to-line current angle.
- $P = \sqrt{3}V_L I_L \cos \theta$ is the **total**. Per phase is one third of it.
- Per-phase method: convert Δ loads with $\mathbf{Z}_\Delta/3$ first.
- b is -120° , c is $+120^\circ$ in abc sequence.

Exam-format traps

Draw the per-phase circuit

Sketch the single-phase equivalent (source V_p , line $\mathbf{Z}$, $\mathbf{Z}_Y$). It shows the method and prevents $\sqrt{3}$ mistakes.

State which quantity

Write “line” or “phase” next to every voltage and current you give.

Summary

Sequence	abc: $\mathbf{V}_{an} = V_p \angle 0^\circ$, $\mathbf{V}_{bn} = V_p \angle -120^\circ$, $\mathbf{V}_{cn} = V_p \angle 120^\circ$
Y	$V_L = \sqrt{3} V_p$ (leads by 30°), $I_L = I_p$
Δ	$V_L = V_p$, $I_L = \sqrt{3} I_p$ (lags by 30°); $\mathbf{Z}_Y = \mathbf{Z}_\Delta / 3$
Power	$P = \sqrt{3} V_L I_L \cos \theta = 3 I_p^2 R$; constant in time
Per phase	solve phase a, shift $\mp 120^\circ$, multiply power by 3
Correction	Q_C total, split by 3; Y caps at V_p , Δ caps at V_L
Grid	high-voltage transmission cuts $I^2 R$ loss; homes get 120/240 V split-phase

Final check

8 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after the last problem.

P1: line voltage

A Y source has $\mathbf{V}_{an} = 277\angle 0^\circ$ V RMS. Find $\mathbf{V}_{ab}$.

your work here ↓

P2: balanced Y–Y

$V_p = 120 \text{ V RMS}$, $\mathbf{Z}_Y = 6 + j8 \Omega$. Find I_L , P , Q and pf.

your work here ↓

P3: balanced Δ

$\mathbf{Z}_{\Delta} = 12 + j9 \, \Omega$ on a 208 V (line) system. Find I_p , I_L and P .

your work here ↓

P4: conversion

Convert $\mathbf{Z}_{\Delta} = 30\angle 45^{\circ} \Omega$ to its Y equivalent.

your work here ↓

P5: motor current

A 50 kW three-phase motor at pf 0.8 lagging runs on 600 V (line). Find I_L .

your work here ↓

P6: correction

Correct the motor of P5 to pf 0.95 lagging (60 Hz). Find Q_C and the capacitance of each Δ -connected capacitor.

your work here ↓

P7: transmission

5 MW is sent over a line of $2\ \Omega$ total resistance (single-phase view).
Compare the loss at 20 kV and 200 kV.

your work here ↓

P8: exam style

A balanced Y load, $\mathbf{Z}_Y = 16 + j12 \Omega$, is on a 240 V (line) system. Find I_L , pf, P and Q .

your work here ↓

Answer sheet

Full worked solutions to P1–P8.

P1: solution

Rule Y: line voltage is $\sqrt{3}$ times the phase voltage and 30° ahead

Size $\sqrt{3} \times 277 = 479.8 \approx 480 \text{ V}$

Result angle $0^\circ + 30^\circ$: $\mathbf{V_{ab} = 480\angle 30^\circ \text{ V}}$ (a standard 277/480 V system)

P2: solution

Per phase $|\mathbf{Z}_Y| = \sqrt{36 + 64} = 10$: $I_L = I_p = 120/10 =$ 12 A

P $3I^2R = 3 \times 144 \times 6 =$ 2592 W

Q $3I^2X = 3 \times 144 \times 8 =$ 3456 var

pf $R/|\mathbf{Z}| = 6/10 =$ 0.6 lagging

P3: solution

I_p Δ : each impedance has the full line voltage:

$$208/|12 + j9| = 208/15 = 13.9 \text{ A}$$

I_L Δ line current: $\sqrt{3} \times 13.87 = 24.0 \text{ A}$

P $3I_p^2 R = 3 \times 13.87^2 \times 12 = 6922 \text{ W}$

P4: solution

Rule balanced: $\mathbf{Z}_Y = \mathbf{Z}_\Delta/3$ (divide the size, keep the angle so the pf is unchanged)

Result $30/3 = 10$: $10\angle 45^\circ \Omega$

P5: solution

Formula $P = \sqrt{3} V_L I_L \text{ pf} \Rightarrow I_L = \frac{P}{\sqrt{3} V_L \text{ pf}}$

Result $\frac{50\,000}{\sqrt{3} \times 600 \times 0.8} = \frac{50\,000}{831.4} = 60.1 \text{ A}$

P6: solution

Angles $\cos^{-1} 0.8 = 36.87^\circ$ ($\tan = 0.75$), $\cos^{-1} 0.95 = 18.19^\circ$
($\tan = 0.329$)

Q_C $50(0.75 - 0.329) = 21.1 \text{ kvar}$ total, so $21\,066/3 = 7022$
var per capacitor

$C (\Delta)$ each sees the line voltage: $\frac{7022}{(377)(600^2)} = 51.7 \mu\text{F}$

P7: solution

20 kV $I = 5 \times 10^6 / 20 \times 10^3 = 250 \text{ A}$, $\text{loss} = 250^2 \times 2 = 125 \text{ kW}$

200 kV $I = 25 \text{ A}$, $\text{loss} = 25^2 \times 2 = 1.25 \text{ kW}$

Ratio 10 times the voltage $\Rightarrow$ 10 times less current $\Rightarrow$ 100 times less loss

P8: solution

V_p Y on a 240 V (line) system: $240/\sqrt{3} = 138.6$ V per phase

$$I_L \quad |Z_Y| = \sqrt{256 + 144} = 20: \quad I_L = 138.6/20 = 6.93 \text{ A}$$

$$\text{pf} \quad 16/20 = 0.8 \text{ lagging}$$

$$P \quad 3 \times 6.93^2 \times 16 = 2304 \text{ W}$$

$$Q \quad 3 \times 6.93^2 \times 12 = 1728 \text{ var}$$